

# Alessio Amaolo

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## EDUCATION

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- Princeton University** – Ph.D. Chemistry and Materials. Advised in Electrical and Computer Engineering. Expected 2027
- Princeton University (3.9/4.0 GPA)** – M.A. Chemistry (Pass with distinction) May 2023
- Selected coursework: Mathematical Methods of Engineering, Numerical Algorithms for Scientific Computing (Python, C++), Nanophotonics (MPB/Scheme), Quantum Optics, Condensed Matter Physics, Quantum Field Theory
- California Institute of Technology (3.9/4.0 GPA)** BSc Chemistry June 2021
- Selected graduate coursework: Computational Physics (Python), Quantum Computation, Advanced Condensed Matter Physics, Advanced Inorganic Chemistry
  - Selected undergraduate coursework: Computing Systems (C, Assembly), Programming Methods (C++), Bayesian Statistics (R)

## PUBLICATIONS – [Google Scholar](#) (\* indicates equal contribution)

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13. Meiburg, A., *et al.* (including **Amaolo, A.**). The Bound-Beating Benchmark: A continuous, lean-verified benchmark of open quantitative problems. In preparation.
12. **Amaolo, A.**, Molesky, S., Rodriguez, A.W., Johnson, S.G., Chao, P. The photonic gearbox: ultra-high Purcell enhancement via symmetry breaking. In preparation.
11. Chen, F.J., **Amaolo, A.**, Chao, P., Molesky, S., Lin, Z., Rodriguez, A.W. [Optically Incoherent Photonic Mutual Information](#). *arXiv:2607.13153*. 2026.
10. **Amaolo, A.**, Chen, F., Chao, P., Maldonado, T.J., Strekha, B., Molesky, S., Rodriguez, A.W. Optimal inverse design in nanophotonics: a tutorial. *Advances in Optics and Photonics (Invited review paper)*. In preparation.
9. Virally, P., Chao, P., **Amaolo, A.**, Rodriguez, A.W., Molesky, S. [How many channels can a photonic system support?](#) *arXiv:2510.01128*. 2025.
8. Molesky, S., Chao, P., **Amaolo, A.**, Rodriguez, A.W. [Inferring structure via duality for photonic inverse design](#). *arXiv:2504.14083*. 2025.
7. Chao, P.\*, **Amaolo, A.\***, Molesky, S., Rodriguez, A.W. [Bounds as blueprints: towards optimal and accelerated photonic inverse design](#). *Optics Express*. 2026.
6. **Amaolo, A.\***, Chao, P.\*, Strekha, B., Clarke, S., Mohajan, J., Molesky, S., Rodriguez, A.W. [Maximum Shannon capacity of photonic structures](#). *npj Nanophotonics*. 2026.
5. Strekha, B., **Amaolo, A.**, Mohajan, J., Chao, P., Molesky, S., Rodriguez, A.W. [Limitations on bandwidth-integrated passive cloaking](#). *Physical Review A*. 2024.
4. **Amaolo, A.**, Chao, P., Maldonado, T.J., Molesky, S., Rodriguez, A.W. [Physical limits on Raman scattering: the critical role of pump and signal co-design](#). *Physical Review A Letters*. 2024.
3. Maldonado, T.J., Pham, D.N., **Amaolo, A.**, Rodriguez, A.W., Türeci, H.E. [Negative electrohydrostatic pressure between superconducting bodies](#). *Physical Review B*. 2024.
2. **Amaolo, A.**, Chao, P., Maldonado, T.J., Molesky, S., Rodriguez, A.W. [Can photonic heterostructures provably outperform single-material geometries?](#) *Nanophotonics*. 2024.
1. McNicholas, B.J., Nie, C., Jose, A., Oyala, P.H., Takase, M.K., Henling, L.M., Barth, A.T., **Amaolo, A.**, Hadt, R.G., Solomon, E.I., Winkler, J.R., Gray, H.B., Despagne-Ayoub, E. [Boronated Cyanometallates](#). *Inorg. Chem.* 2022.

## RESEARCH EXPERIENCE

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### Student Researcher, Google DeepMind

Research on computational lithography and mask process correction for chip fabrication optimization.

Summer  
2025

### Graduate Researcher in Rodriguez Group, Princeton

Performing research at the intersection of convex optimization and photonics focusing on calculating fundamental limits on photonic device performance.

2021-  
Present

- Developed novel algorithms for photonic inverse design utilizing convex relaxations, beating state-of-the-art algorithms 10x in performance in 10x fewer iterations.
- Formulated first information-theoretic treatment of photonic devices to calculate optimal electromagnetic information transfer, including in antennas and waveguides (e.g., fiber-optics).
- Extended methods of calculating performance limits for photonic multi-material, non-linear optical, and cloaking devices.
- Developed an open-source package for calculating photonic performance limits. [GitHub repo.](#)
- Currently working on heat transfer formulations for limits on broadband thermophotovoltaics.

### Undergraduate Research Fellow, Chan Group, Caltech

Researched tensor networks for simulating many-body quantum physics.

Summer  
2020

### Undergraduate Research Fellow, Gray Group, Caltech

Researched the ligand field theory and electronic structure of boronated cyanometallates via spectroscopic methods.

Summer  
2019

### Undergraduate Research Fellow, Gray Group, Caltech

Researched methods of electrochemical reduction of carbon dioxide to produce oxygen.

Summer  
2018

## CONFERENCE TALKS

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20th International Congress on Artificial Materials for Novel Wave Phenomena – Invited Talk  
“Communicating through waves: How many optical channels can a photonic device support?”

2026

Photonics North – Invited Talk  
“Certificates of Suboptimality & Opportunities for New Photonic Designs”

2026

16th International Conference on Metamaterials, Photonic Crystals and Plasmonics – Invited Talk  
“The photonic gearbox: ultra-high local density of states via cylindrical-symmetry breaking”

2026

## JOURNAL REVIEWING

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*Journal of the Optical Society of America B*  
*Computational and Applied Mathematics*

*ACS Photonics*  
*Science Advances*

## TECHNOLOGIES

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Proficient in Python (+NumPy, SciPy), scientific computing and ML in JAX, C/C++, MPI, OpenMP.

## AWARDS

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Princeton Chemistry Department David V. Milligan '62 Fellowship

2021-2022

- Awarded by faculty in recognition of academic excellence

Reed and Ruth Brantley Research Fellowship

Summer 2020

Doris S. Perpall SURF Speaking Competition semifinalist (top 10% of presentations)

2019

Donald Voet and Jerome Vinograd Research Fellowship

Summer 2019

Dr. and Mrs. Daniel C. Harris Research Fellowship

Summer 2018

## TEACHING

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| Teaching Assistant, <b>Electronic Circuit Design, Analysis and Implementation</b> – Held two weekly labs on electronic circuits. <a href="#">Teaching reviews.</a>                                                        | Spring 2026                |
| Teaching Assistant, <b>Robot Planning Meets Machine Learning</b> – Held help sessions and graded theory and programming questions for a graduate course on planning in robotics. Reviews hidden (new course).             | Fall 2025                  |
| Teaching Assistant, <b>Optical and Quantum Electronics</b> – Held help sessions and wrote questions for a graduate course on light-matter interactions for applications in electronics. <a href="#">Teaching reviews.</a> | Fall 2024                  |
| Teaching Assistant, <b>Electromagnetism for Engineers</b> – Held two weekly labs focusing on applications of electromagnetism to engineering. <a href="#">Teaching reviews.</a>                                           | Spring 2024<br>Spring 2025 |
| Teaching Assistant, <b>Ecology</b> – Held three weekly labs focusing on experimental techniques in ecology. Prepared midterm and final exam questions. <a href="#">Teaching reviews.</a>                                  | Fall 2023                  |
| Teaching Assistant, <b>Electromagnetism</b> – Held regular problem-solving sessions, office hours, and labs for a first-year advanced course on electromagnetism. <a href="#">Teaching reviews.</a>                       | Spring 2023                |
| Teaching Assistant, <b>General Chemistry</b> – Held three weekly recitations with a mix of lecturing and problem solving. Prepared midterm and final exam questions. <a href="#">Teaching reviews.</a>                    | Fall 2022                  |
| Teaching Assistant, <b>Statistical Mechanics</b> – Held weekly recitations, office hours, and an exam review session for an undergraduate course in statistical mechanics. <a href="#">Teaching reviews.</a>              | Spring 2021                |
| Teaching Assistant, <b>Physical Chemistry</b> – Held weekly recitations and graded problem sets for a course on quantum chemistry. Prepared <a href="#">notes</a> for the second part of the course.                      | Fall 2019                  |

### Selected Student Teaching Feedback

- *Electromagnetism for Engineers*: “Alessio was extremely helpful, cared about our understanding of the material, and was a very fair and kind individual.”
- *Statistical Mechanics*: “Alessio was an amazing TA. He was always very kind in the way he answered questions without making me feel like it was a dumb question. One of the best physics TAs I've encountered.”
- *General Chemistry*: “I feel as if most of my understanding of the course came from Alessio and his inherent ability to teach a group of students effectively.”